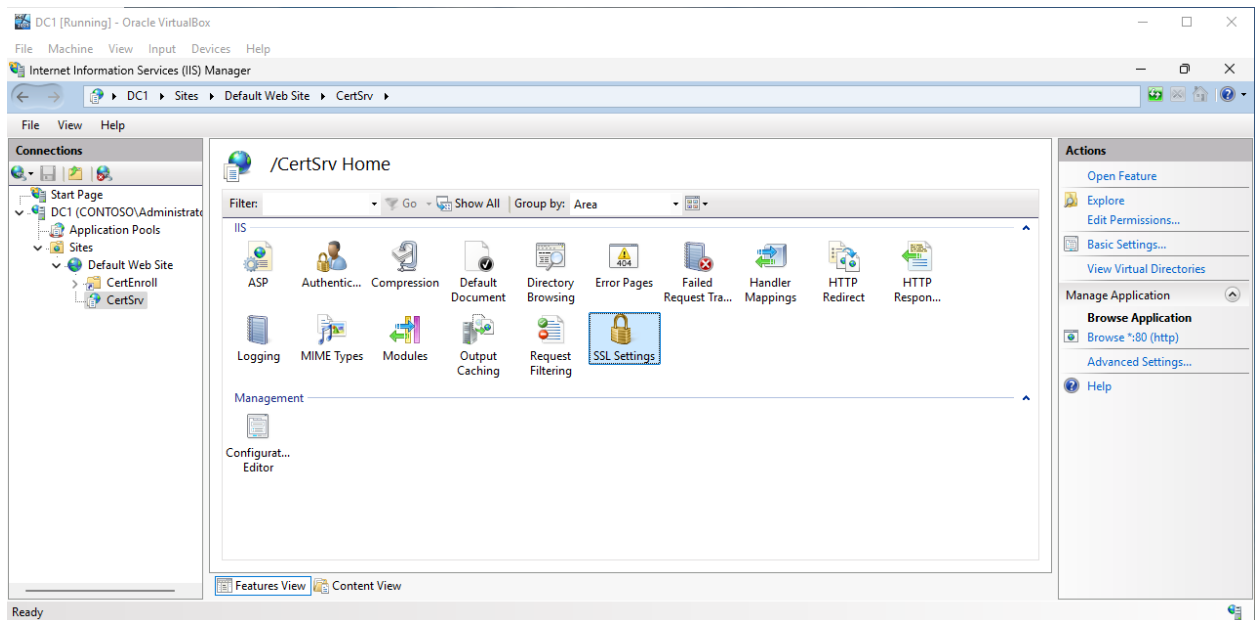
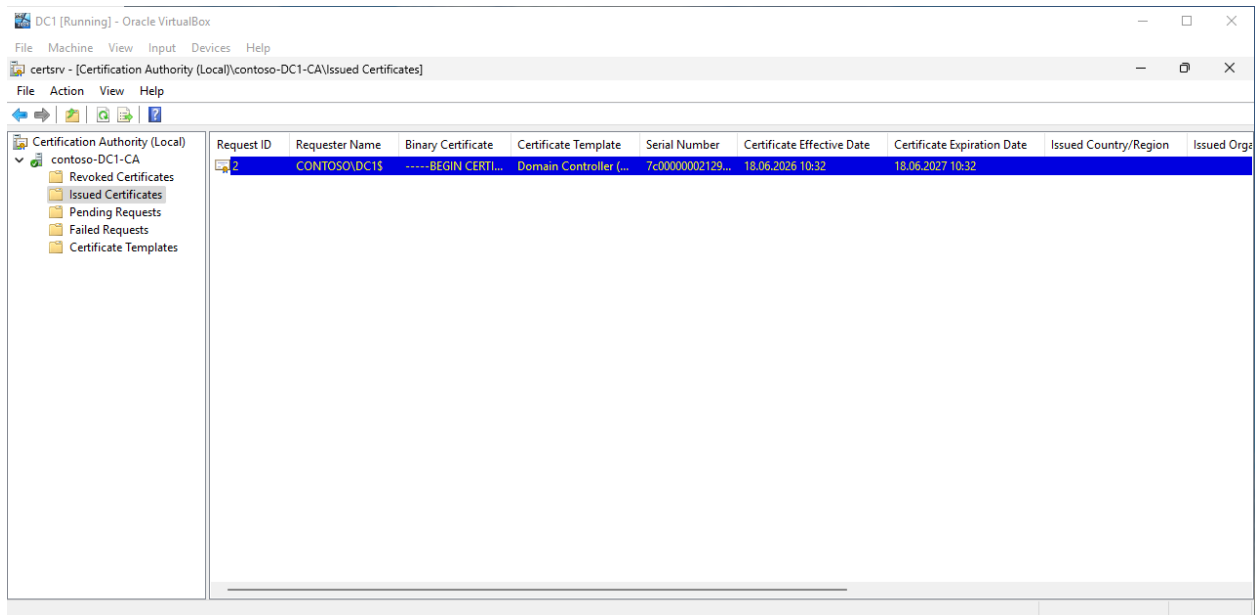


Lab 14 – My Solutions

1: Exercise 1. Deploy Enterprise Root CA:



2: Exercise 2. Windows Firewall — Allow HTTP via Group Policy:

```
CL1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Administrator: PowerShell
PowerShell 7.6.0
PS C:\Users\Administrator.CONTOSO> Test-NetConnection dc1.contoso.internal -Port 80

ComputerName : dc1.contoso.internal
RemoteAddress : 192.168.56.200
RemotePort : 80
InterfaceAlias : Ethernet
SourceAddress : 192.168.56.101
TcpTestSucceeded : True

PS C:\Users\Administrator> Get-NetFirewallRule -PolicyStore ActiveStore -DisplayName "Allow HTTP Inbound GPO" | Select-Object DisplayName, Enabled, PolicyStoreSourceType

DisplayName      Enabled PolicyStoreSourceType
-----
Allow HTTP Inbound GPO      True      GroupPolicy

PS C:\Users\Administrator.CONTOSO> Test-NetConnection -ComputerName dc1.contoso.internal -Port 80

ComputerName : dc1.contoso.internal
RemoteAddress : 192.168.56.200
RemotePort : 80
InterfaceAlias : Ethernet
SourceAddress : 192.168.56.101
TcpTestSucceeded : True
```

3: Exercise 3. Capture Cleartext Credentials with Wire-shark:

The screenshot shows a web browser window on the left displaying the 'Microsoft Active Directory Certificate Services' welcome page. The page includes a 'Welcome' section, a paragraph about requesting certificates, a link to download certificates, and a 'Select a task' section with links for requesting, viewing status, and downloading certificates. On the right, the Wireshark network traffic capture window is open, showing a list of captured packets. The selected packet is an HTTP GET request for '/certsrv/certsrv' from 192.168.56.101 to 192.168.56.200. The packet details pane shows the request line: 'GET /certsrv HTTP/1.1'.

No.	Time	Source	Destination	Protocol	Length	Info
56	41.782294200	192.168.56.101	192.168.56.200	HTTP	536	GET /certsrv HTTP/1.1
58	41.783880600	192.168.56.200	192.168.56.101	HTTP	82	HTTP/1.1 401 Unauthorized (text/html)
80	55.034406400	192.168.56.101	192.168.56.200	HTTP	599	GET /certsrv HTTP/1.1
81	55.051191400	192.168.56.200	192.168.56.101	HTTP	421	HTTP/1.1 301 Moved Permanently (text/html)
82	55.055465200	192.168.56.101	192.168.56.200	HTTP	600	GET /certsrv/ HTTP/1.1
87	55.497035200	192.168.56.200	192.168.56.101	HTTP	1039	HTTP/1.1 200 OK (text/html)
89	55.524180800	192.168.56.101	192.168.56.200	HTTP	586	GET /certsrv/certsrv.gif HTTP/1.1
90	55.526023400	192.168.56.200	192.168.56.101	HTTP	362	HTTP/1.1 200 OK (GIF89a)

CL1 [Running] - Oracle VirtualBox

FileMachineViewInputDevicesHelp

Microsoft Active Directory Certificate Services

Microsoft Active Directory Certificate Services

Welcome

Use this Web site to request a certificate mail client, or other program. By using a your identity to people you communicate and encrypt messages, and, depending t you request, perform other security tasks

You can also use this Web site to downlo (CA) certificate, certificate chain, or certifi or to view the status of a pending request

For more information about Active Direct see [Active Directory Certificate Services](#).

Select a task:
[Request a certificate](#)
[View the status of a pending certificate](#)
[Download a CA certificate, certificate chain](#)

Ethernet

FileEditViewGoCaptureAnalyzeStatisticsTelephonyWirelessToolsHelp

Not securedc1.conto...

http

No.	Time	Source	Destination	Protocol	Length	Info
56	41.782294200	192.168.56.101	192.168.56.200	HTTP	536	GET /certsrv HTTP/1.1
58	41.783880600	192.168.56.200	192.168.56.101	HTTP	82	HTTP/1.1 401 Unauthorized (text/html)
80	55.034406400	192.168.56.101	192.168.56.200	HTTP	599	GET /certsrv HTTP/1.1
81	55.051191400	192.168.56.200	192.168.56.101	HTTP	421	HTTP/1.1 301 Moved Permanently (text/html)
82	55.055465200	192.168.56.101	192.168.56.200	HTTP	600	GET /certsrv/ HTTP/1.1
87	55.497035200	192.168.56.200	192.168.56.101	HTTP	1039	HTTP/1.1 200 OK (text/html)
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90	55.526023400	192.168.56.200	192.168.56.101	HTTP	362	HTTP/1.1 200 OK (GIF89a)

Sequence Number (raw): 1502174957
[Next Sequence Number: 1574 (relative sequence number)]
Acknowledgment Number: 1856 (relative ack number)
Acknowledgment number (raw): 1113217842
0101 = Header Length: 20 bytes (5)
> Flags: 0x018 (PSH, ACK)
Window: 254
[Calculated window size: 65024]
[Window size scaling factor: 256]
Checksum: 0xf4ba [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
> [Timestamps]
> [SEQ/ACK analysis]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
TCP payload (546 bytes)
▼ Hypertext Transfer Protocol
> GET /certsrv/ HTTP/1.1\r\nHost: dc1.contoso.internal\r\nConnection: keep-alive\r\nCache-Control: max-age=0\r\n▼ Authorization: Basic Y29udG9zb1xhZG1pbm1zdHJhdG9yO1BhY3N3M9HJk\r\nCredentials: contoso\administrator:P@ssw0rd
Upgrade-Insecure-Requests: 1\r\nUser-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/
Accept-Encoding: gzip, deflate\r\nAccept-Language: en-US,en;q=0.9\r\n\r\n[Response in frame: 87]
[Full request URI: http://dc1.contoso.internal/certsrv/]

0000 63 6f 6e 74 6f 73 6f 5c 61 64 6d 69 6e 69 73 74 contoso

0010 72 61 74 6f 72 3a 50 40 73 73 77 30 72 64 rator:

Packet (600 bytes) Basic Credentials (30 bytes)
Packets: 98 · Displayed: 8 (8.2%) · Dropped: 0 (0.0%) Profile: Default

4: Exercise 4. Protect HTTP Traffic with IPsec:

The image displays two Oracle VM VirtualBox windows side-by-side, both running Windows Server 2016. The top window is titled 'DC1 [Running] - Oracle VirtualBox' and the bottom window is titled 'CL1 [Running] - Oracle VirtualBox'. Both windows show the 'certlm - [Certificates - Local Computer\Personal\Certificates]' console window.

DC1 [Running] - Oracle VirtualBox

certlm - [Certificates - Local Computer\Personal\Certificates]

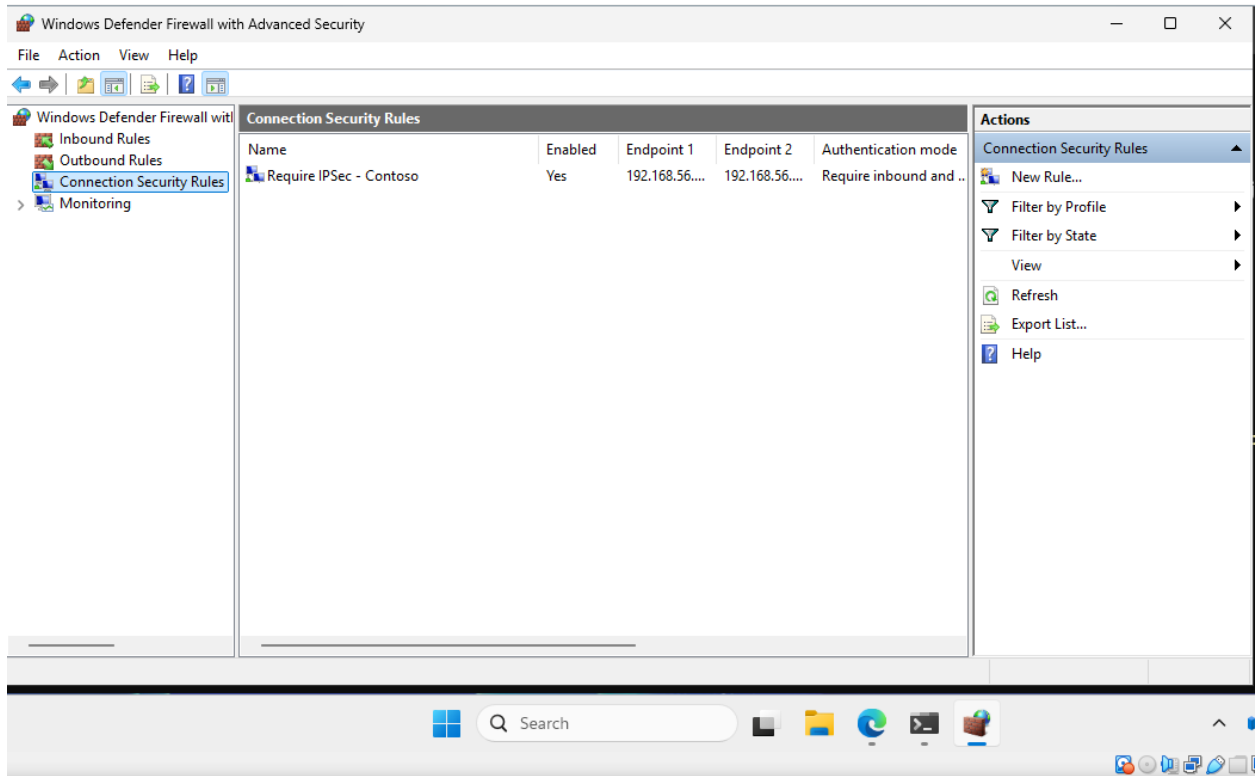
Issued To	Issued By	Expiration Date	Intended Purposes	Friendly Name	Status	Certificate Te...
contoso-DC1-CA	contoso-DC1-CA	18.06.2031	<All>	<None>		
dc1.contoso.internal	contoso-DC1-CA	18.06.2027	Client Authenticati...	<None>		Domain Contr...
dc1.contoso.internal	contoso-DC1-CA	19.06.2027	Client Authenticati...	<None>		Domain Contr...

Personal store contains 3 certificates.

CL1 [Running] - Oracle VirtualBox

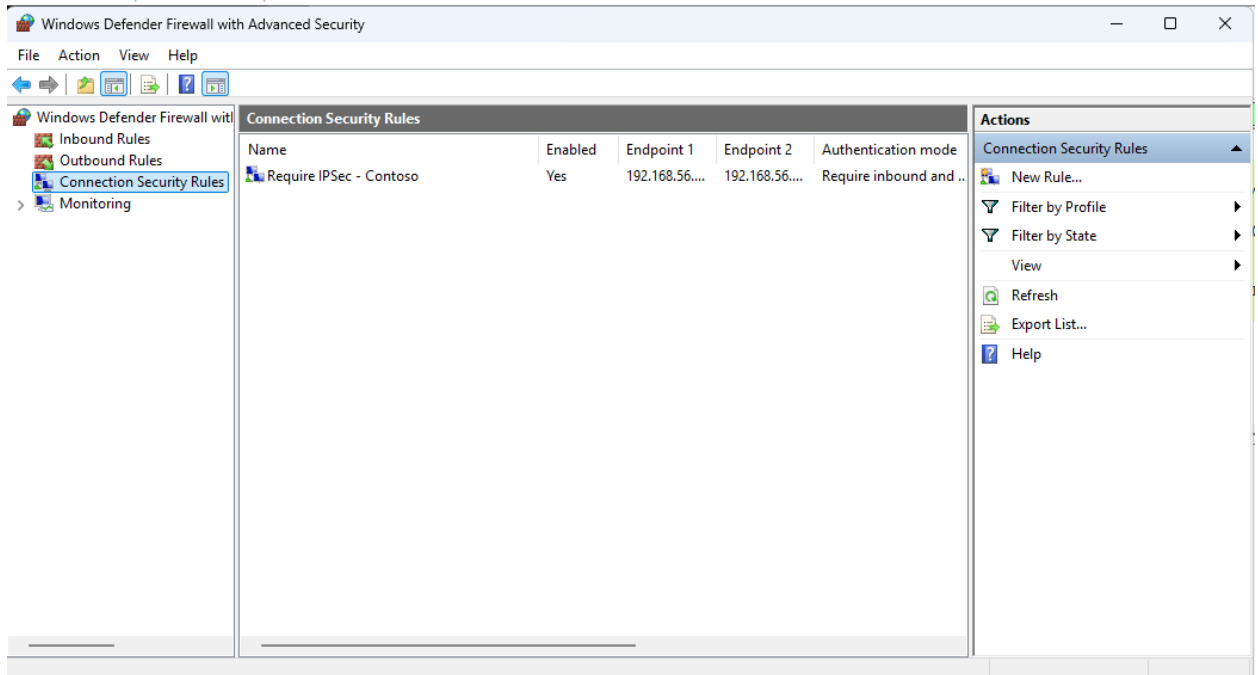
certlm - [Certificates - Local Computer\Personal\Certificates]

Issued To	Issued By	Expiration Date	Intended Purposes	Friendly Name	Status	Certificate Te...
cl1.contoso.internal	contoso-DC1-CA	6/19/2027	Client Authenticati...	<None>		Computer



[Running] - Oracle VirtualBox

Machine View Input Devices Help



CL1 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Microsoft Active Directory Certificate Services - X

Not secure dc1.contoso.internal

Capturing from Ethernet

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

esp

No.	Time	Source	Destination	Protocol	Length	Info
134	19.308991900	192.168.56.200	192.168.56.101	ESP	78	ESP (SPI=0x58db3ac2)
135	19.308994400	192.168.56.200	192.168.56.101	ESP	78	ESP (SPI=0x58db3ac2)
136	19.309203300	192.168.56.101	192.168.56.200	ESP	78	ESP (SPI=0x0c1f3150)
138	19.650422400	192.168.56.101	192.168.56.200	ESP	102	ESP (SPI=0x0c1f3150)
139	19.661453400	192.168.56.200	192.168.56.101	ESP	202	ESP (SPI=0x58db3ac2)
145	32.157354600	192.168.56.101	192.168.56.200	ESP	78	ESP (SPI=0x0c1f3150)
146	32.158194300	192.168.56.200	192.168.56.101	ESP	90	ESP (SPI=0x58db3ac2)
159	47.658222100	192.168.56.101	192.168.56.200	ESP	78	ESP (SPI=0x0c1f3150)
160	47.659517600	192.168.56.200	192.168.56.101	ESP	90	ESP (SPI=0x58db3ac2)
162	48.374276800	192.168.56.101	192.168.56.200	ESP	106	ESP (SPI=0x0c1f3150)
163	48.374295000	192.168.56.101	192.168.56.200	ESP	106	ESP (SPI=0x0c1f3150)
164	48.386159600	192.168.56.200	192.168.56.101	ESP	218	ESP (SPI=0x58db3ac2)
165	48.386160600	192.168.56.200	192.168.56.101	ESP	218	ESP (SPI=0x58db3ac2)

Frame 3: Packet, 678 bytes on wire (5424 bits), 678 bytes captured (5184 bits) on interface 0

Section number: 1

> Interface id: 0 (\Device\NPF_{247034C7-527A-4B10-8C2E-2D8D4D2D5F2}

Encapsulation type: Ethernet (1)

Arrival Time: Jun 19, 2026 08:37:40.616746700 Central European Day

UTC Arrival Time: Jun 19, 2026 06:37:40.616746700 UTC

Epoch Arrival Time: 1781851060.616746700

[Time shift for this packet: 0.000000000 seconds]

[Time delta from previous captured frame: 372.039500 milliseconds]

[Time since reference or first frame: 2.487015500 seconds]

Frame Number: 3

Frame Length: 678 bytes (5424 bits)

CL1 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Microsoft Active Directory Certificate Services - X

Not secure dc1.contoso.internal

Capturing from Ethernet

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

No.	Time	Source	Destination	Protocol	Length	Info
-----	------	--------	-------------	----------	--------	------